

REVIT

CONCRETE

Concrete structural modeling in Autodesk Revit. The track case study: twenty modules from datum setup to printed sheets.

20

MODULES IN FULL

72

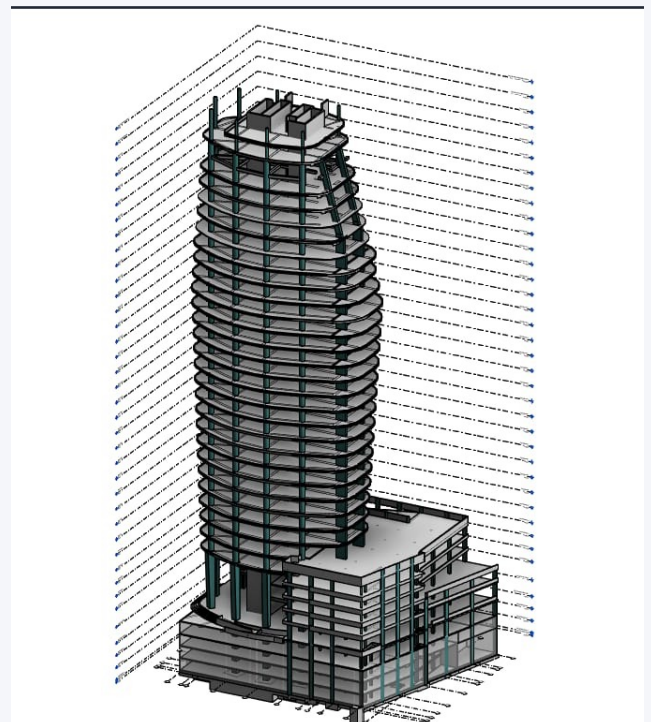
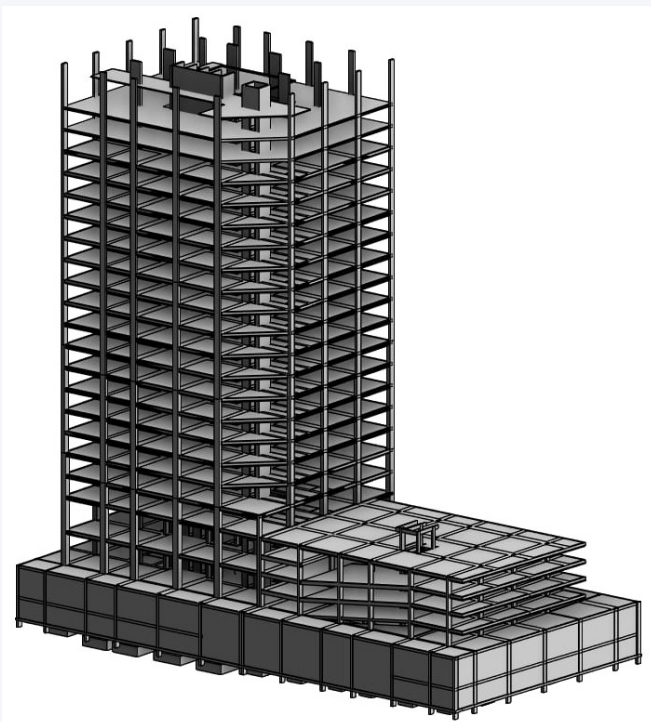
SUBTOPICS

4 wks

PER TRACK

100%

LIVE ONLINE



01 | CONTEXT

BACKGROUND

Background

Revit Concrete Structure Training is a live instructor led track within the Structra Revit for Structures program. Over four weekday evening weeks you move from the Revit interface, levels and grids through columns, foundations, beams, slabs, staircases and roofs, then into reinforcement placement, detailing, bar bending schedules and construction sheets.

The track is 100% practical and built for engineering students and early career structural professionals who want to produce real, buildable reinforced concrete models.

START DATE

Monday, 14
September 2026,
third week of
September

FORMAT

100% online, live
sessions led by an
instructor on Zoom

DURATION

4 weeks per track,
weekday evenings

FEE

KSH 2,500 per track

WHO IT'S FOR

University
engineering students
and early career
structural and BIM
professionals

PREREQUISITE

A laptop capable of
running Revit, no
prior Revit experience
required

02 | BEFORE TRAINING

THE CHALLENGE

The challenge

01

Who it is for

University engineering students and early career structural and BIM professionals who want to build real, buildable models, not just pretty renders.

02

Starting level

No prior Revit experience is required. The only prerequisite is a laptop capable of running Autodesk Revit.

03

Format and pace

100% online, live sessions led by an instructor on Zoom, weekday evenings, four weeks per track.

04

The gap it closes

Moving from knowing Revit exists to modeling reinforced concrete structures end to end, with details, schedules and sheets.

03 | CURRICULUM

THE SOLUTION

The solution, module by module

All 20 modules in full, 72 practical subtopics, covered in the live sessions.

01 Brief overview of Revit (what it entails)

Difference with AutoCAD software; Benefits over AutoCAD software.

02 Setting up the software

Brief introduction of Libraries required; Downloading other extensions from Autodesk site.

03 Beginning with Revit

Creating a new project; An overview of the templates; An overview of palettes, tabs and panels.

04 Creating Datum Elements

Creating Levels; Creating grids , (same as with the levels creation); Propagate options extents on the grid; Customizing and changing the shape and appearance of grid bubbles; Linking CAD files to create grids.

05 Brief Overview of Creating 3D elements

Foundations, Columns, Beams, Slabs.

06 Creating Columns

Placement of columns; Loading column families.

07 Creating Foundations

Hosting foundations on the columns created; Creating Strip footing; Using eccentricity option; Basement floor slab; Loading foundation families into your drawings.

08 Creating Beams and slabs

Copying beams and slabs from level to level; Creating a slab; Hollow pot and waffle slab.

09 Curved Beams

Procedure of modeling the curved beams.

10 Beam System Layout and Justification Rules

Using the beam system to model the structural beams; Modelling the beams using the beam system layout rules; Modelling the beams using the beam system Justification Rules; Deselecting the beam system.

11 Creating Staircase and Ramps

Types of staircase; Creating an architectural ramp and structural ramp; Modeling Staircase as a ramp; Shaft openings.

12 Visibility/Graphics and Filters

Applying Visibility/Graphic Overrides , Method 1; Applying Visibility/Graphic Overrides , Method 2; Applying Visibility/Graphic Overrides , Method 3; Applying Visibility/Graphic Overrides , Method 4.

13 Creating Roofs and Truss placement

Sloped and gable ended roofing; Adding system trusses on your roof; Editing and creating your own customized trusses.

14 Placing of Reinforcement

Creating sections in your drawing; Foundation Reinforcement; Column Reinforcement; Beam Reinforcement; Slab reinforcement; Staircase reinforcement; Ramp reinforcement.

15 Annotations

Adding dimensions; Adding spot elevations; Adding spot coordinates; Adding spot slope.

16 Detailing of Sections

Placing breaklines in the sections; Using hatch files to create regions; Placement of rebar tags on your reinforcements; Adding comments on your reinforcement; Tagging of the structural elements.

17 Schedules and Material Takeoffs

Creating Bar Bending Schedules; Loading shape images into bar bending schedules; Creating Material Takeoffs for the structural elements; Sorting and filtering fields in material takeoffs and bar bending schedules.

18 Placing Drawings on Sheets

Loading different paper sizes for your title blocks; Using system loaded title blocks; Editing and customizing the title blocks; Adjusting texts and fields in your title block; Saving title blocks as a Revit family template; Introduction to duplicate options for drawings; Placing drawings on the sheets.

19 Printing Sheets to PDF format

Setting up your sheets before printing; Printing one sheet at a time; Printing several sheets at a time.

20 Conversion of your files into CAD and other formats

Export options for your drawing into other formats; Transporting your drawing to AutoCAD; Sending your model for analysis.

04 | OUTCOME

THE RESULT

The result, what you will be able to build

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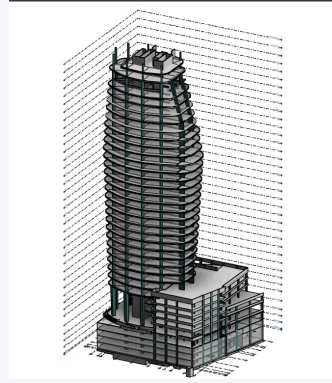
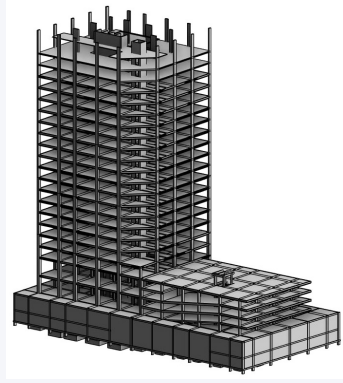
MODULES

72

SUBTOPICS

4 wks

DURATION



Learners complete the track able to model reinforced concrete frames from datum to documentation, with reinforcement, bar bending schedules and printed sheets. The structures shown above are modeled during the program.

REVIT CONCRETE STRUCTURE TRAINING, AUTODESK REVIT

STRUCTRA

ENROL, NEXT COHORT 14 SEPT 2026

structratraining.co.ke · +254 7XX XXX XXX · Start 14 September 2026 · KSH 2,500 per track